

线性代数B 2024 fall期中

1

求下列行向量构成的向量组的秩和一个极大线性无关组：
 $\alpha_1 = (-1, 5, 3, 2), \alpha_2 = (4, 1, -2, 9), \alpha_3 = (2, 0, 1, 4), \alpha_4 = (0, 3, 4, -5)$

2

4. 下述齐次线性方程组何时会有非零解？何时只有零解？

$$\begin{cases} x_1 - 3x_2 - 5x_3 = 0, \\ 2x_1 - 7x_2 - 4x_3 = 0, \\ 4x_1 - 9x_2 + ax_3 = 0, \\ 5x_1 + bx_2 - 55x_3 = 0. \end{cases}$$

3

求解：

$$\begin{vmatrix} 1 & 1 & 1 & \dots & 1 \\ 1 & 2 & 0 & \dots & 0 \\ 1 & 0 & 3 & \dots & 0 \\ \vdots & \vdots & \vdots & & \vdots \\ \vdots & \vdots & \vdots & & \vdots \\ 1 & 0 & 0 & \dots & n \end{vmatrix}$$

1、
$$\begin{bmatrix} -1 & 5 & 3 & 2 \\ 4 & 1 & -2 & 9 \\ 2 & 0 & 1 & 4 \\ 0 & 3 & 4 & -5 \end{bmatrix} \rightarrow \begin{bmatrix} -1 & 5 & 3 & 2 \\ 0 & 21 & 10 & 17 \\ 0 & 10 & 7 & 8 \\ 0 & 3 & 4 & -5 \end{bmatrix} \rightarrow \begin{bmatrix} -1 & 5 & 3 & 2 \\ 0 & 3 & 4 & -5 \\ 0 & 10 & 7 & 8 \\ 0 & 0 & -9 & 26 \end{bmatrix}$$

$$\begin{vmatrix} -1 & 5 & 3 & 2 \\ 0 & 3 & 4 & -5 \\ 0 & 10 & 7 & 8 \\ 0 & 0 & -9 & 26 \end{vmatrix} = \begin{vmatrix} 3 & 4 & -5 \\ 10 & 7 & 8 \\ 0 & -9 & 26 \end{vmatrix} = 3 \cdot \begin{vmatrix} 7 & 8 \\ -9 & 26 \end{vmatrix} - 10 \cdot \begin{vmatrix} 4 & -5 \\ -9 & 26 \end{vmatrix}$$

$$= 3 \times (182 + 72) - 10 \times (104 + 45)$$

$$= 3 \times 254 - 10 \times 149 \neq 0$$

$\therefore \text{rank}=4$, 极大线性无关组为 $\alpha_1, \alpha_2, \alpha_3, \alpha_4$

2、
$$\begin{vmatrix} 1 & -3 & -5 \\ 2 & -7 & -4 \\ 4 & -9 & a \\ 5 & b & -35 \end{vmatrix} = \begin{vmatrix} 1 & -3 & -5 \\ 0 & -1 & 6 \\ 0 & 3 & 20+a \\ 0 & b+15 & -30 \end{vmatrix} = \begin{vmatrix} 1 & -3 & -5 \\ 0 & -1 & 6 \\ 0 & 0 & 38+a \\ 0 & 0 & b+10 \end{vmatrix}$$

若 $a = -38$ 且 $b = -10$, 则有非零解

若 $a \neq -38$ 或 $b \neq -10$, 则只有零解

① 记 D_n

$$\begin{vmatrix} 1 & 1 & 1 & \dots & 1 \\ 1 & 2 & 0 & \dots & 0 \\ 1 & 0 & 3 & \dots & 0 \\ \vdots & \vdots & \vdots & \ddots & \vdots \\ 1 & 0 & 0 & \dots & n \end{vmatrix}$$

$$\therefore D_n = n \cdot D_{n-1} + (-1)^{n+1} \begin{vmatrix} 1 & 2 & 0 & \dots & 0 \\ 1 & 0 & 3 & \dots & 0 \\ \vdots & \vdots & \vdots & \ddots & \vdots \\ 1 & 0 & 0 & \dots & n-1 \\ 1 & 0 & 0 & \dots & 0 \end{vmatrix}$$

$$= n D_{n-1} - (n-1)!$$

$$= n [(n-1) D_{n-2} - (n-2)!] - (n-1)!$$

$$= n \cdot (n-1) [(n-2) D_{n-2} - (n-3)!] - n \cdot (n-2)!$$

$$= n! - \sum_{k=2}^n \frac{n!}{k} \quad (n \geq 2) = n! \left(2 - \sum_{k=1}^n \frac{1}{k} \right)$$

$$n=1 \text{ 时 } D_1 = 1$$

4

已知

$$\begin{vmatrix} x & y & z & x+y+z \\ 3 & 0 & 2 & 0 \\ 1 & 1 & 1 & 1 \\ 2 & 2 & 2 & 1 \end{vmatrix} = \frac{1}{2}$$

求

$$\begin{vmatrix} x-y & y & z-x & x+y+z \\ 3 & 0 & -1 & 0 \\ y-x & 2-y & x-z & 2-x-y-z \\ 3 & 2 & -1 & 1 \end{vmatrix}$$

$$\begin{vmatrix} x-y & y & z-x & x+y+z \\ 3 & 0 & -1 & 0 \\ y-x & 2-y & x-z & 2-x-y-z \\ 3 & 2 & -1 & 1 \end{vmatrix} = \begin{vmatrix} x & y & z & x+y+z \\ 3 & 0 & 2 & 0 \\ 2-x & 2-y & 2-z & 2-x-y-z \\ 5 & 2 & 4 & 1 \end{vmatrix} = \begin{vmatrix} x & y & z & x+y+z \\ 3 & 0 & 2 & 0 \\ 2 & 2 & 2 & 2 \\ 2 & 2 & 2 & 1 \end{vmatrix}$$

$$= 2 \times \frac{1}{2} = 1$$

证明:

$$\text{rank} \begin{pmatrix} A & 0 \\ C & B \end{pmatrix} \geq \text{rank}(A) + \text{rank}(B)$$

记 $\text{rank}(A) = r$, $\text{rank}(B) = s$

$$\therefore A, B \text{ 经过初等行变换后可变为: } A' = \begin{bmatrix} r_1 \\ r_2 \\ \vdots \\ r_r \\ 0 \\ 0 \end{bmatrix} \quad B' = \begin{bmatrix} s_1 \\ s_2 \\ \vdots \\ s_s \\ 0 \\ 0 \end{bmatrix}$$

$$\therefore \text{rank} \begin{pmatrix} A & 0 \\ 0 & B \end{pmatrix} = \text{rank} \begin{bmatrix} A' & 0 \\ 0 & 0 \\ 0 & B' \end{bmatrix} = \text{rank} \begin{bmatrix} A' & 0 \\ 0 & B' \end{bmatrix} = r+s$$

而记 C_1 为 C 经过... 的简化阶梯形矩阵

则 $\begin{pmatrix} A & 0 \\ C & B \end{pmatrix}$ 有子矩阵 $\begin{pmatrix} A' & 0 \\ C_1 & B' \end{pmatrix}$

$$\begin{vmatrix} A' & 0 \\ C_1 & B' \end{vmatrix} = |A'| \cdot |B'| \neq 0,$$

$$\therefore \text{rank} \begin{pmatrix} A & 0 \\ C & B \end{pmatrix} \geq \text{rank} \begin{pmatrix} A' & 0 \\ C_1 & B' \end{pmatrix} = r+s$$

6

β 能由 $\{\alpha_1, \dots, \alpha_s\}$ 线性表出, 但是不能由 $\{\alpha_1, \dots, \alpha_{s-1}\}$ 线性表出, 求证:

$$\text{rank}\{\alpha_1, \dots, \alpha_s\} = \text{rank}\{\alpha_1, \dots, \alpha_{s-1}, \beta\}$$

$$\beta = k_1 \alpha_1 + k_2 \alpha_2 + \dots + k_s \alpha_s$$

由于 β 不能由 $\alpha_1, \alpha_2, \dots, \alpha_{s-1}$ 表出, $\therefore k_s \neq 0$

$$\therefore \alpha_s = \frac{1}{k_s} \beta - \frac{k_1}{k_s} \alpha_1 - \frac{k_2}{k_s} \alpha_2 - \dots - \frac{k_{s-1}}{k_s} \alpha_{s-1}$$

$\therefore \alpha_s$ 可以由 $\alpha_1, \alpha_2, \dots, \alpha_{s-1}, \beta$ 线性表出

$\therefore \alpha_1, \alpha_2, \dots, \alpha_s$ 与 $\alpha_1, \dots, \alpha_{s-1}, \beta$ 等价

$$\therefore \text{rank}\{\alpha_1, \alpha_2, \dots, \alpha_s\} = \text{rank}\{\alpha_1, \alpha_2, \dots, \alpha_{s-1}, \beta\}$$

7

$$\begin{bmatrix} 1 & 1 & 2 & x \\ 3 & 2 & 1 & y \\ 1 & -1 & 0 & z \end{bmatrix}$$

的行向量组何时与

$$\begin{bmatrix} 3 & 1 & 5 & 14 \\ 0 & 2 & -1 & 11 \\ 0 & 0 & 3 & 3 \end{bmatrix}$$

$$x_1(3, 1, 5, 4) + x_2(0, 2, -1, 11) + x_3(0, 0, 3, 3) = (1, 1, 2, x)$$

等价?

$$\alpha_1 = \begin{bmatrix} 3 \\ 1 \\ 5 \\ 14 \end{bmatrix}$$

$$\alpha_2 = \begin{bmatrix} 0 \\ 2 \\ -1 \\ 11 \end{bmatrix}$$

$$\alpha_3 = \begin{bmatrix} 0 \\ 0 \\ 3 \\ 3 \end{bmatrix}$$

$$\therefore \begin{vmatrix} 3 & 0 & 0 \\ 1 & 2 & 0 \\ 5 & -1 & 3 \\ 14 & 11 & 3 \end{vmatrix} = 0, \quad \text{rank} = 3,$$

$$\text{rank}(\alpha_1, \alpha_2, \alpha_3, \beta_i) \geq 3,$$

$$\text{令} \begin{vmatrix} 3 & 0 & 0 & 1 \\ 1 & 2 & 0 & 1 \\ 5 & -1 & 1 & 2 \\ 14 & 11 & 1 & x \end{vmatrix} = \begin{vmatrix} 1 & 2 & 0 & 1 \\ 0 & 3 & 0 & 1 \\ 0 & -11 & 1 & -3 \\ 0 & -17 & 1 & x-14 \end{vmatrix} = \begin{vmatrix} 1 & 2 & 0 & 1 \\ 0 & 3 & 0 & 1 \\ 0 & -11 & 1 & -3 \\ 0 & 6 & 0 & 11-x \end{vmatrix}$$

$$= 0 \quad \therefore 11-x = 2, \quad x = 9.$$

$$\begin{vmatrix} 3 & 0 & 0 & 3 \\ 1 & 2 & 0 & 2 \\ 5 & -1 & 1 & 1 \\ 14 & 11 & 1 & y \end{vmatrix} = 3 \cdot \begin{vmatrix} 1 & 0 & 0 & 1 \\ 0 & 2 & 0 & 1 \\ 0 & -1 & 1 & -4 \\ 0 & 11 & 1 & y-14 \end{vmatrix}$$

$$= 3 \cdot \begin{vmatrix} 2 & 0 & 1 \\ -1 & 1 & -4 \\ 11 & 1 & y-14 \end{vmatrix} = \begin{vmatrix} -1 & 1 & -4 \\ 0 & 2 & -2-7 \\ 0 & 12 & y-58 \end{vmatrix}$$

$$\therefore y-58 = -42, \quad y = \cancel{44} \quad 16$$

$$\begin{vmatrix} 3 & 0 & 0 & 1 \\ 1 & 2 & 0 & -1 \\ 5 & -1 & 1 & 0 \\ 14 & 11 & 1 & z \end{vmatrix} = \begin{vmatrix} 1 & 2 & 0 & -1 \\ 0 & -6 & 0 & 4 \\ 0 & -11 & 1 & 5 \\ 0 & -17 & 1 & z+14 \end{vmatrix} = \begin{vmatrix} 1 & 2 & 0 & -1 \\ 0 & -3 & 0 & 2 \\ 0 & -11 & 1 & 5 \\ 0 & -6 & 0 & z+9 \end{vmatrix}$$

$$\therefore z+9 = 4, \quad z = -5$$

$$\therefore \begin{cases} x = 9 \\ y = \cancel{44} \quad 16 \\ z = -5 \end{cases}$$

例3 设 n 个方程的 n 元齐次线性方程组的系数矩阵 A 的行列式等于 0, 并且 A 的 (k, l) 元的代数余子式 $A_{kl} \neq 0$ 。证明:

$$\eta = \begin{pmatrix} A_{k1} \\ A_{k2} \\ \vdots \\ A_{kn} \end{pmatrix}$$

是这个齐次线性方程组的一个基础解系。

$$\text{当 } m \neq k \text{ 时, } \sum_{i=1}^n a_{mi} A_{ki} = 0$$

$$\therefore |A| = \sum_{i=1}^n a_{ki} A_{ki} = 0,$$

$$\therefore \forall m, \sum_{i=1}^n a_{mi} A_{ki} = 0$$

$$\therefore A\vec{\eta} = 0 \quad \text{又有 } \vec{\eta} \neq 0 \quad \therefore \vec{\eta} \text{ 是方程的一个非零解}$$

$$\text{而 } |A| = 0, \text{ 又有 } A_{kl} \neq 0, \therefore \text{rank}(A) = n-1, \text{ 解空间维数为 } 1.$$

$$\therefore \vec{\eta} \text{ 是一个基础解系}$$

已知方程组:

$$a_{11}x_1 + \dots + a_{1n}x_n = 0$$

...

$$a_{n1}x_1 + \dots + a_{nn}x_n = 0$$

的解都是方程

$$b_1x_1 + \dots + b_nx_n = 0$$

的解, 求证:

$\beta = \{b_1, \dots, b_n\}$ 可以由 $\{\alpha_1, \dots, \alpha_n\}$ 线性表出, 其中 α_i 为系数矩阵 A 的行向量。

β 可由 $\alpha_1, \alpha_2, \dots, \alpha_n$ 线性表出, 即

$$\alpha_1 x_1 + \alpha_2 x_2 + \dots + \alpha_n x_n = \beta \text{ 有解}$$

即:

$$\begin{cases} a_{11}x_1 + a_{21}x_2 + \dots + a_{n1}x_n = b_1, \\ a_{12}x_1 + a_{22}x_2 + \dots + a_{n2}x_n = b_2 \\ \vdots \\ a_{1n}x_1 + a_{2n}x_2 + \dots + a_{nn}x_n = b_n \end{cases} \quad \text{有解}$$

若 β 不能由 $\alpha_1, \alpha_2, \dots, \alpha_n$ 线性表出

则有 $\text{rank}\{\alpha_1, \alpha_2, \dots, \alpha_n, \beta\} = \text{rank}\{\alpha_1, \alpha_2, \dots, \alpha_n\} + 1$

记 $\text{rank}\{\alpha_1', \alpha_2', \alpha_3', \dots, \alpha_n'\} = r$, $\text{rank}\{\alpha_1, \alpha_2, \dots, \alpha_n, \beta\} = r+1$

则方程组的解空间为 $n-r$ 维

设方程组的解 $\eta = k_1\eta_1 + k_2\eta_2 + \dots + k_{n-r}\eta_{n-r}$

由于方程组的解都是 $\beta'x = 0$ 的解

$$\therefore (\alpha_1', \alpha_2', \dots, \alpha_n', \beta') \cdot \begin{bmatrix} x_1 \\ x_2 \\ \vdots \\ x_n \end{bmatrix} = 0 \text{ 与原方程组同解}$$

$$\therefore \text{rank}\{\alpha_1', \alpha_2', \dots, \alpha_n', \beta'\} = r$$

与已知矛盾

$$\therefore \text{rank}\{\alpha_1, \alpha_2, \dots, \alpha_n, \beta\} = \text{rank}\{\alpha_1, \alpha_2, \dots, \alpha_n\}$$

即 β 可由 $\alpha_1, \alpha_2, \dots, \alpha_n$ 线性表出